

James Howard

EDUCATION

B.S. Degree: Electrical Engineering

9/1995 - 6/1998

COCERTIFICATION

AWS Certified Solutions Architect – Professional

May 2020.

SUMMARY

I am a U.S. citizen with nearly 20 years of experience in advanced engineering

- 18 years (including my PhD) in Artificial Intelligence AI encompassing advanced machine learning (ML), deep learning (DL), convolutional neural networks (CNN, NLP, LLM, Generative AI etc.

Throughout my career, **I have worked with some of the most prestigious companies in Silicon Valley and beyond, including Lockheed Martin, Space Systems Loral, Microsoft (where I served four times as a Senior Solution Architect for various multibillion-dollar clients), Amazon, Hewlett-Packard, Texas Instruments, Ericsson, T-Mobile, AT&T, Medtronic, Ernst & Young, Johnson & Johnson, Navy Federal Credit Union, Optum, UnitedHealthcare, Humana, Convey Health, Prudential Financial.**

I have led the development of numerous systems, guiding them from initial concept and architecture through to successful commercial deployment.

OBJECTIVE

Primarily seeking consulting and contract opportunities, but also open to permanent positions.

EXPERIENCE

Microsoft – Client Sites West Coast to East Coast

9/2024-Present

Senior AI/ML Architect | Principal GenAI & Agentic AI Engineer | Hands-on Developer

Hands-on role architecting, building, and deploying **enterprise-scale GenAI, Agentic AI**, AI-powered surveillance, and intelligent automation platforms across **Azure, AWS, and GCP** for Microsoft and regulated Fortune 500 customers. Designed and delivered production-grade AI systems including **LLM-powered email and communications monitoring**, AI copilots and virtual assistants (**Microsoft Copilot, ChatGPT Enterprise, Gemini Advanced**), **RAG / GraphRAG** document intelligence, **real-time behavioral anomaly detection and market abuse surveillance (SIEM / RegTech)**, **AI observability and LLMops dashboards**, and autonomous coding and DevOps agents (**GitHub Copilot, CodeWhisperer**). Led deployments aligned with **Responsible AI**, explainability, auditability, and **regulatory compliance** requirements in financial services, healthcare, and government environments.

Architected **distributed agentic systems and autonomous AI workflows** using **LangGraph, AutoGen, CrewAI 2.0, Semantic Kernel, DSPy**, and actor-based concurrency frameworks (Erlang/Elixir) enabling **tool use, planning, memory, self-reflection (Reflexion), Tree/Graph-of-Thought, ReAct**, and multi-agent coordination. Built interoperability layers using **MCP (Model Context Protocol)** and **ACP (Agent Control Protocol)** for **secure context sharing, policy enforcement, human-in-the-loop (HITL)**, and cross-model orchestration. Implemented **GraphRAG / GraphRAG++ and Hybrid RAG architectures** combining **knowledge graphs** (Neo4j, TigerGraph, ArangoDB, NebulaGraph) with **vector + keyword retrieval** to maximize reasoning accuracy, grounding, and autonomous decision-making.

- **Multi-cloud GenAI platforms:** Built **enterprise GenAI stacks** using **Azure OpenAI, Amazon Bedrock, Google Vertex AI, Anthropic, Mistral, Cohere**, and open-source LLMs; integrated **Copilot Studio, Azure AI Studio, Vertex**

Agent Builder, LlamaStack with Infrastructure as Code (Terraform, Pulumi, CDK, Bicep) and CI/CD pipelines (GitHub Actions, GitLab CI/CD, CodeCatalyst).

• **Models (frontier + OSS): GPT-4/4.1/4o/4.2, Claude 3.5 Sonnet/Opus, Gemini 1.5 Pro/Flash, Llama 3/3.1, Mistral Large/Mixtral, DBRX, Cohere Command-R+, Qwen 2.5, Phi-4; multimodal GPT-4V, Claude Vision, LLaVA 1.6, BLIP-2, WhisperX.**

Amazon, Seattle, Washington

3/2024-9/2024

Senior AI/ML Architect | Hands-on Developer & Engineer

Worked on the Alexa next generation, architected, created the necessary DevOps cloud infrastructure and developed (hands-on) the data pipeline and the AI/ML GenAI system to facilitate IoT hardware monitoring and data anomaly detection for the new generation of Amazon Alexa system.

Designed and deployed advanced Generative AI (GenAI) and Retrieval-Augmented Generation (RAG) solutions on AWS using Amazon Bedrock, SageMaker, and OpenSearch with cutting-edge foundation models including GPT-4, Claude 3, Mistral/Mixtral, Cohere Command R+, Titan, and LLaMA 3. Architected scalable RAG pipelines leveraging vector databases (Pinecone, FAISS, OpenSearch), semantic search, and embedding generation with SageMaker JumpStart and LangChain. Integrated AWS services such as Lambda, API Gateway, DynamoDB, and Step Functions to orchestrate GenAI workflows with secure, real-time document retrieval and context-aware LLM interactions. Built multi-agent reasoning systems using CrewAI and LangGraph, enabling complex task automation and dynamic memory management. Implemented prompt engineering, safety guardrails, and model evaluation with PromptFlow, Traceloop, and Ragas. Delivered enterprise-grade GenAI platforms with secure access control (IAM, KMS), logging (CloudWatch), and CI/CD automation using CodePipeline and Terraform.

AT&T, Dallas Texas

2/2022-3/2024

Senior AI/ML Architect | Hands-on Developer & Engineer

At AT&T, served as a Senior AI/ML Architect but as a hands-on senior developer writing code, creating cloud infrastructure DevOps, participating in testing and deployment, I led, designed and deployed Generative AI and RAG-powered solutions on AWS to support 5G anomaly detection, predictive maintenance, and intelligent customer service automation.

Architected and deployed Retrieval-Augmented Generation (RAG) systems using FAISS, Dense Passage Retrieval (DPR), Siamese BERT, OpenSearch, and Amazon Kendra, enabling context-aware Q&A and dynamic support insights by fusing dense vector retrieval with transformer-based response generation.

Integrated state-of-the-art LLMs including GPT-3, GPT-Neo, BERT, T5, RoBERTa, DistilBERT, XLNet, ELECTRA, ALBERT, CTRL, and GPT-J using Hugging Face Transformers, SageMaker JumpStart, and OpenAI API, to power summarization, log understanding, chatbot response synthesis, and dynamic document generation.

Microsoft, New York, Newport Beach California

3/2019-2/2022

Senior AI/ML Architect | Hands-on Developer & Engineer

As a Solution Architect, high-level lead, and expert. I collaborated with numerous prominent Microsoft clients, primarily Fortune 500 companies, to enhance their existing architectures or design entirely new ones. My role encompassed guiding clients through cutting-edge technologies and implementing effective solutions. Below are some of the key areas I have worked in:

As a Senior AI/ML Architect | Hands-on Developer & Engineer, I designed, architected, and led enterprise-scale AI initiatives across on-prem, AWS, Azure, and GCP environments, delivering real-time credit auditing, CRM automation, intelligent email classification, AI auto-reply, and anomaly detection systems. Leveraging Convolutional Neural Networks (CNNs), transformer-based models, and multi-agent systems, I deployed scalable Generative AI and Retrieval-Augmented Generation (RAG) architectures powered by GPT-2, GPT-3, T5, BERT, RoBERTa, DistilBERT, XLNet, ELECTRA, and ALBERT. I implemented RAG pipelines using FAISS, Haystack, OpenSearch, Hugging Face Transformers, and Amazon Kendra, enabling semantic search, document summarization, and contextual Q&A. I integrated embeddings from OpenAI, Sentence-BERT, and Universal Sentence Encoder to support deep similarity search and hybrid retrieval. GenAI deployments were unified using SageMaker, Azure ML, and Vertex AI, orchestrated via LangChain, FastAPI, and Airflow, and scaled with Kubernetes, Docker, Terraform, and Kubeflow for CI/CD and model serving. Optimized transformer inference using NVIDIA V100/A100 GPUs, TensorRT, ONNX Runtime, and DeepSpeed, while applying Low-Rank Adaptation (LoRA) and knowledge distillation for model compression. Enabled multi-agent collaboration through early frameworks like Haystack pipelines and custom agent routing, and integrated Explainable AI (XAI) via SHAP and LIME. Applied RLHF, Chain-of-Thought prompting, and ethical AI design patterns to ensure transparency, safety, and alignment with responsible AI principles.

In AWS, Lead, architected and developed a comprehensive AWS infrastructure with: Redshift (also tested POC in Snowflake Data Warehouse and Snowpipe), RDS, EMR, MSK (Kafka Confluent), IoT (Internet of things), SCADA, PLC, S3 AWS Compute E2C, Glue, AWS Connect, DMS, Athena, EC2, RDS Aurora MySQL PostgreSQL, Couchbase, MemSQL, Elasticsearch, Lambda projects with full Big Data, Amazon Elastic MapReduce (EMR), CI/CD (CICD, CI CD, CI-CD) pipeline, CloudFormation CFN, Spark, Kafka Confluent, Oozie, Sqoop, NiFi, Pig, Hive, Hbase, MSK, AWS CLI, Amazon EMR File System (EMRFS). Comprehensive CDK and CloudFormation experience with writing script in TypeScript, JavaScript, Python, Java, and C#. Collaborative notebooks Zeppelin, Jupyter. AWS networking used VPC, Private Subnet, Public Subnet, Internet Gateway, Routing Security groups etc. Visualization and analysis used QuickSight and CloudSearch. Also created a Kubernetes cluster using Elastic Container Service for Kubernetes (EKS), App Mesh, EC2 Container Service (ECS), Fargate. For AI used deep learning frameworks like Apache MXNet Machine Learning and Deep Learning development and deployment, VMware, AWS Developer Tools, AWS Management Tools, Amazon Machine Learning, SageMaker, Alexa Skills Kit, AWS DeepLens, Amazon Deep Learning AMIs, Amazon TensorFlow on AWS, Amazon CodeWhisperer AI and other components. For security used Identity and Access Management (IAM), AWS Organizations, Multi-Factor Authentication, SAS, migration of On-Prem Windows SAS to AWS SAS Studio etc. Also lead, architected and helped developing multiple CDK and CloudFormation projects with writing script in TypeScript, JavaScript, Python and Java.

In Azure, designed and built an end-to-end enterprise pipeline using Azure Data Factory Gen2 to orchestrate ingestion from SQL, flat files, Apptio Cloudability, data warehouses, and external sources. Architected a 4-zone ADLS Gen2 data lake (Raw, Staging, Curated, Consumption) aligned with Microsoft's best practices. Streamed critical data via Kafka Confluent and less critical via Azure Event Hub into the Raw Zone. Integrated Delta Lake, supporting table versioning, schema evolution, and ACID transactions. Configured Azure Databricks clusters with Unity Catalog for workspace-level governance, access control, lineage, and auditing. Managed model lifecycles using MLflow, Unity Catalog, and REST API endpoints for deployment and tracking. Built real-time and web apps with .NET, Angular, React.js, Spring Boot, and Node.js, deployed inside and outside AKS with APIM and Service Mesh. Designed dashboards in Power BI, integrating with Synapse Analytics, Microsoft Fabric Lakehouses,

OneLake, and Azure Purview for metadata and cataloging. Enhanced cloud security with Microsoft Sentinel, Azure AD, AADDS, Key Vault, Service Principals, Private Links, and Azure VNet. Developed LLM-powered solutions using Azure OpenAI, ChatGPT API, Prompt Engineering, and Microsoft Bot Framework for sentiment analysis and automation. Implemented CI/CD pipelines using Azure DevOps, GitHub Actions, Azure Boards, and Azure Application Insights. Automated infra-as-code with ARM Templates, Terraform, and Visual Studio. Designed data warehousing in Azure Synapse and Snowflake, with optimized pipelines and DWH performance tuning. Enabled HIPAA-compliant FHIR/HL7 integration using FHIR APIs, REST services, and JSON for EHR interoperability. Delivered full-stack observability and analytics using Highcharts, Encore Analytics Empower, and custom telemetry dashboards.

In GCP, worked on different component of the GCP among others Cloud Dataproc, BigQuery, Airflow Cloud Dataflow, Cloud Data Fusion, Cloud Dataprep, Data Catalog, Google Kubernetes Engine (GKE), Kubernetes CLusters, Dataproc CLusters, Container Registry, CI/CD (CICD, CI CD, CI-CD) pipelines, Deep Learning Containers, Cloud Bigtable, Cloud SQL, Firebase Realtime Database. For artificial intelligence used AI building blocks, Text-to-Speech, Speech-to-Text, AutoML, Vision AI, Vertex AI, Cloud Natural Language, Video AI, AI Platform, AI Hub and AI Platform Deep Learning VM Image etc.

SKILLS

Summary of Technical Expertise

With 20+ years of experience and a PhD from University of California **UCLA/Davis**, I'm a Senior Solution/Enterprise/Technical Architect, Engineering Leader, and hands-on Developer across **AI/ML, GenAI, Cloud, Big Data, Kubernetes, DevOps, Data Warehousing, and Enterprise Systems** (AWS, Azure, GCP, on-prem, hybrid).